Jesse He

Curriculum Vitae

HDSI 432, 3234 Matthews Lane
La Jolla, CA, 92093
(419) 378-5584
jeh020@ucsd.edu
jessehe.inbox@gmail.com
he-jesse.github.io

Education

Halıcıoğlu Data Science Institute, UC San Diego, San Diego, California
PhD Candidate, Advised by Yusu Wang, Gal Mishne

The Ohio State University, Columbus, OH
Bachelor of Science in Mathematics (Honors), *magna cum laude* May 2022
Bachelor of Arts in Computer and Information Science, *magna cum laude* May 2022

Research Interests

My research interests lie in the intersection of interpretable machine learning and geometric data analysis. In particular, I am interested in explainability methods for graph neural networks and interpretable manifold learning.

Employment

Research

SU2024- **National Security Internship, AI and Data Analytics**, *Pacific Northwest National Laboratory*, Richland, WA (remote)
Supervised by Henry Kvinge

SU2023- **Graduate Student Researcher**, *Halıcıoğlu Data Science Institute, UC San Diego*, San Diego, California

SU2021 **Emerging Issues in Cybersecurity REU**, *New Mexico Tech*, Sorocco, NM (remote)
Supervised by Subhasish Mazumdar

Teaching

Graduate Teaching Assistant, *UC San Diego, Halıcıoğlu Data Science Institute*

WI 2024 DSC 206: Algorithms for Data Science

Undergraduate Grader, *The Ohio State University Department of Mathematics*

SP2022 Math 3345H: Honors Foundations of Higher Mathematics

SP2022 Math 5591H/5112: Honors Abstract Algebra II

AU2021 Math 5590H/5111: Honors Abstract Algebra I

Undergraduate Grader, *The Ohio State University Department of Computer Science and Engineering*

AU2020-SP2021 CSE 3521: Survey of Artificial Intelligence I

AU2019 CSE 2221: Software Components

Digital Sandbox Project Group Instructor, *The Ohio State University Media, Marketing, and Communications Scholars*

SP2020 Introduction to LaTeX

Awards

Qualcomm Innovation Fellowship 2024

Publications

He, J., Yusu Wang, and Gal Mishne (2026). "Local Manifold Explanations with Tangent Space Regression". In: *Proceedings of the 2nd Conference on Topology, Algebra,*

and Geometry in Data Science (TAG-DS 2026). URL: https://proceedings.mlr.press/v334/he26a.html.

Jenne, Helen, Davis Brown, **J. He**, Max Vargas, and Henry Kvinge (2026). "Even with AI, Bijection Discovery is Still Hard: The Opportunities and Challenges of OpenEvolve for Novel Bijection Construction". In: *Proceedings of the 2nd Conference on Topology, Algebra, and Geometry in Data Science (TAG-DS 2026)*. (Spotlight). URL: https://proceedings.mlr.press/v334/jenne26a.html.

He, **J.**, Helen Jenne, Max Vargas, Davis Brown, Gal Mishne, Yusu Wang, and Henry Kvinge (2026). "MINAR: Mechanistic Interpretability for Neural Algorithmic Reasoning". (In Submission). arXiv: 2602.21442 [cs.LG]. URL: https://arxiv.org/abs/2602.21442.

Kohli, Dhruv, **J. He**, Chester Holtz, Alexander Cloninger, and Gal Mishne (2026). "Robust boundary detection and density estimation using doubly stochastic scaling of the Gaussian kernel". (In Submission). arXiv: 2411.18942 [math.ST]. URL: https://arxiv.org/abs/2411.18942. Co-first author.

He, **J.**, Akbar Rafiey, Gal Mishne, and Yusu Wang (2025b). "Explaining GNN Explanations with Edge Gradients". In: *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2*. Association for Computing Machinery. DOI: 10.1145/3711896.3736947. URL: https://doi.org/10.1145/3711896.3736947.

He, **J.**, Helen Jenne, Herman Chau, Davis Brown, Mark Raugas, Sara C. Billey, and Henry Kvinge (2025a). "Machines and Mathematical Mutations: Using GNNs to Characterize Quiver Mutation Classes". In: *Proceedings of the 42nd International Conference on Machine Learning*. URL: https://proceedings.mlr.press/v267/he25g.html.

Chau, Herman, Helen Jenne, Davis Brown, **J. He**, Mark Raugas, Sara C. Billey, and Henry Kvinge (2025). "Machine Learning meets Algebraic Combinatorics: A Suite of Datasets Capturing Research-level Conjecturing Ability in Pure Mathematics". In: *Proceedings of the 42nd International Conference on Machine Learning*. (Oral). URL: https://proceedings.mlr.press/v267/chau25a.html.

He, **J.**, Tristan Brugère, and Gal Mishne (2023). "Product Manifold Learning with Independent Coordinate Selection". In: *Proceedings of 2nd Annual Workshop on Topology, Algebra, and Geometry in Machine Learning (TAG-ML)*. URL: https://proceedings.mlr.press/v221/he23a.html.